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SAP Two-Tier ERP Integration Recommendation Guide

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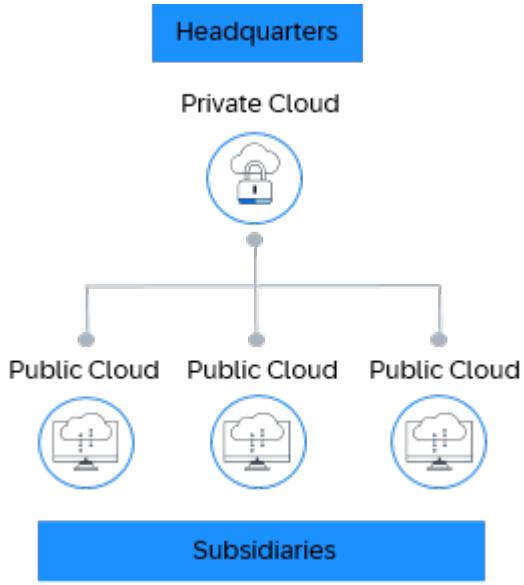
1 SAP Two-Tier ERP

Two-Tier ERP provides an ERP solution for companies with subsidiaries.

Organizations, including their subsidiaries, face technological gaps hindering growth. To modernize and transform, they need to update their entire technology portfolio, digitalize operations, and consider all applications, deployment types, integration aspects, and innovation opportunities. To enhance business value, enterprises should simplify their business and become more innovative.

Two-Tier ERP comprises complex and customizable ERP for the headquarters, as well as ERP for the subsidiaries that supports quick scalability with low landscape complexity and costs.

The headquarters and subsidiaries ERP systems are integrated in a hybrid landscape with suitable implementation and process optimizations, as represented in the following image:



SAP Two-Tier ERP Integration

Identify and address integration requirements early in the process. A comprehensive understanding of end-to-end business processes, including both master data and transactional data flows is crucial. SAP S/4HANA Cloud also needs to be integrated with various other solutions within the customer's system landscape.

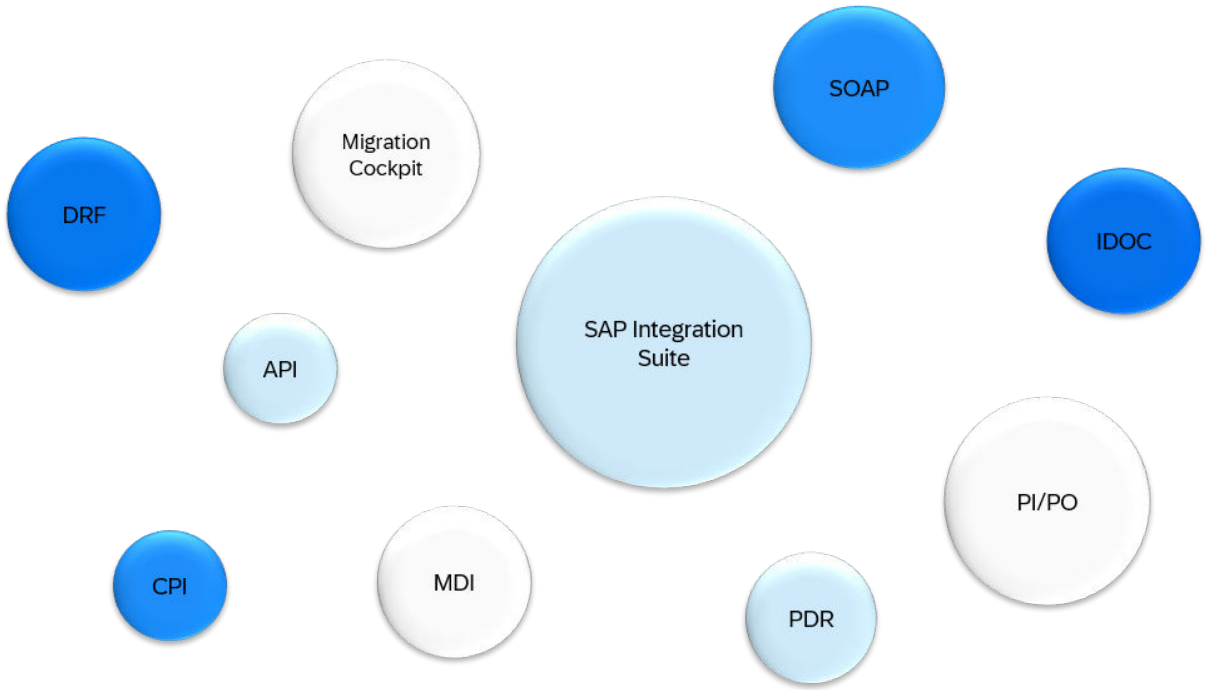
Integration Challenges

Two-Tier ERP implementation needs to address the following challenges to integration:

- **Complex and fragmented customer landscapes**
  These landscapes are typically characterized by a mix of legacy systems, cloud-based solutions, on-premise software, and third-party tools that are operating independently or with minimal integration.
- **Multiple integration options**

While having many options may seem advantageous, it can present several difficulties such as complexity in decision making, compatibility issues, technical expertise, resource allocation and scalability.

- **Complex and time-consuming decision-making process**
  Owing to an overwhelming number of options, businesses often require thorough analysis, testing, and stakeholder alignment to ensure they arrive at the most suitable integration solution that serves both immediate and long-term needs.



Multiple Integration Options

2 About This Guide

This guide aims to provide customers and partners with a detailed and user-friendly overview that helps them thoroughly understand various integration solutions available within the SAP ecosystem. It can serve as an essential resource for navigating the complexities of integration, offering tailored insights and strategic recommendations. By doing so, it ensures that our customers and partners are equipped to choose the optimal solution based on the specific attributes of their IT landscape, business needs, and long-term growth objectives.

Specifically, this guide aims to provide the following:

- **Comprehensive understanding of the available options**
  - The guide provides a detailed overview of all the available SAP integration frameworks and tools, such as SAP Cloud Platform Integration (SAP CPI), SAP Master Data Integration (SAP MDI), SAP API Management, and SAP Business Technology Platform (SAP BTP).
  - Each solution is thoroughly explained, with a special accent on its core features, use cases, technical requirements, and how it fits into different IT environments (cloud, on-premise, or hybrid).
- **Tailored recommendations based on specific landscapes**
  - The guide emphasizes that the most suitable integration solution depends on the customer's unique IT landscape. It offers tools and methods to assess the current landscape, including legacy systems, cloud environments, and hybrid setups.
- **Decision-making framework**
  - The guide provides a structured decision-making framework, helping customers and partners align their integration needs with the best available solutions.

3 Master Data Replication

Master data is the core data that remains the same over a long period of time. It contains information about business entities, products, and locations, which makes it essential for running operations within an organization or a business unit.

Replicating master data across headquarters and their subsidiaries is important for the following reasons:

- **Data consistency**– Ensures that both headquarters and subsidiaries have access to the same up-to-date information, and thus, minimizing discrepancies and errors in data.
- **Operational efficiency**– Streamlines processes and operations by maintaining synchronized data; therefore, reducing the need for manual data entry and reconciliation.
- **Regulatory compliance**– Helps in maintaining compliance with regulatory requirements by ensuring that all parts of the organization adhere to the same standards and data practices.

There are two options for replicating your master data:

- Master Data Integration [page 6]
- Data Replication Framework [page 13]

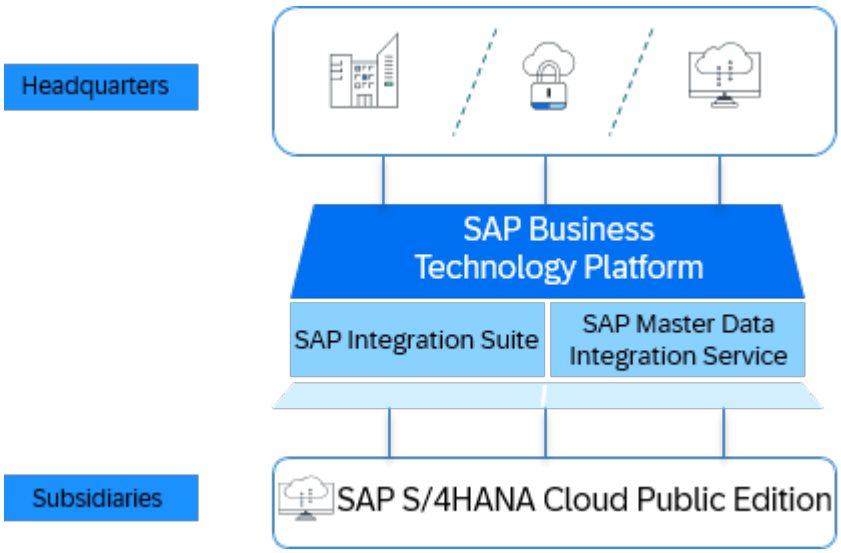
Choose the option that best suits your system landscape, and business requirements, among other factors.

3.1 Master Data Integration

Master Data Integration (MDI) is one of two solutions for replicating your master data in Two-Tier ERP.

Consider opting for MDI in the following cases:

- Your headquarters is in private cloud, while its multiple subsidiaries are in public cloud.
- You are using other SAP cloud solutions or non-SAP solutions that needed to be integrated.
- Harmonizing master data across your system landscape is a requirement.
- You have a SAP Business Technology Platform account.

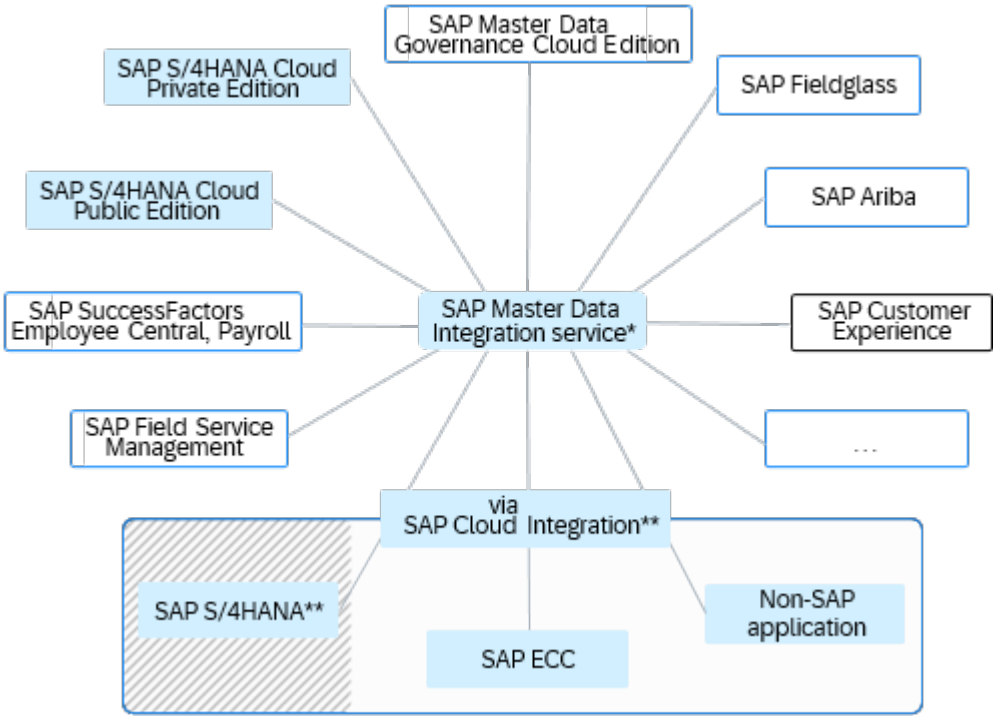


3.1.1 SAP Master Data Integration Service

This cloud service enables you to handle distributed and federated sets of master data in a heterogeneous landscape. Thus, SAP MDI is the central point for integrating master data with the SAP Integration Suite.

The service uses SAP One Domain Model as the exchange format to achieve a consistent view of master data across end-to-end business processes in the integrated SAP Intelligent Suite.

SAP Master Data Integration provides reliable real-time replication in a hub-and-spoke approach. This allows every business application to have its own copy of the relevant master data.



* - Free of charge, ready-to-use integration, standard for SAP-branded cloud applications

** - Utilize SAP Integration Suite when ready-to-use integration is not available

... - Current integration of CX solutions mostly via SAP Integration Suite (direct integration on roadmap)

The outlined boxes indicate the areas that fall outside the scope of Two-Tier ERP, while the blue shaded boxes signify the areas that are within the scope of Two-Tier ERP.

3.1.2 SAP Master Data Integration Capabilities

The SAP MDI service provides the following capabilities:

- **Distribution of master data**– You can specify a distribution model that allows you to identify which systems have access to view or modify specific data.
- **Data editing by multiple systems**– You can specify a distribution model that allows you to identify which systems have access to view or modify specific data.
- **Read access logging**– You can monitor and log read access to sensitive data for data protection and privacy purposes.
- **Extensibility**– You can extend the aligned business model with fields and entities to support business requirements.
- **SOAP-REST interoperability**– Business partner and product master data can be replicated via SOAP interfaces and ODM-compliant REST interfaces that act in an interoperable fashion.

3.1.3 Guiding Principles of the SAP Master Data Integration Service

Two models explain how the SAP MDI Service works. Implementation according to these models helps simplify complexity and eliminate manual data work.

Central Integration Point

MDI follows the central integration point, which is often called a hub-and-spoke model. In this model, MDI acts as the centralized system (hub) to manage data exchanges between multiple systems (spokes). Instead of connecting systems directly to each other, having a hub comes with centralized management, which reduces complexity and facilitates scalability and error handling.

One Domain Model

The SAP One Domain Model (ODM) is an integrated data model that standardizes the representation of business objects across the SAP ecosystem. ODM ensures that these business objects are defined consistently across all SAP applications, including SAP S/4HANA, SAP SuccessFactors, SAP Ariba, and others.

ODM decouples from application specifics and provides well defined semantics for business objects.

Manual data mapping by customers is not required any more.

3.2 Master Data Integration Setup Requirements from Two-Tier ERP

When integrating MDI with applications, there are two primary approaches: direct integration, or leveraging the SAP Integration Suite.

Direct integration is feasible when a ready-to-use or out-of-the-box" integration for MDI is available. In scenarios where such prebuilt integrations are not provided, the Integration Suite serves as a bridge to establish the connection between MDI and the target application.

Learn the required setup for specific business objects when you are replicating master data between SAP S/4HANA Cloud Private Edition, MDI, and SAP S/4HANA Cloud Public Edition.

3.2.1 Setup Requirements for Business Partner Object

The following tasks let you replicate the object **Business Partner** from **SAP S/4HANA Cloud Private Edition** to **MDI**, and **MDI** to **SAP S/4HANA Cloud Public Edition**.

For MDI SOAP Inbound (SAP S/4HANA Cloud Private Edition to MDI):

1. Create an MDI service instance for SAP S/4HANA Cloud Private Edition application. For more information, see Connecting Applications via Service Instances.
2. Set up an MDI business system (one-time activity per tenant). For more information, see Configuring Own Business System ID.
3. Set up communication arrangement SAP_COM_0008. For more information, see SAP Note 3087667.
4. Create confirmation destinations on SAP BTP for SAP S/4HANA Cloud Private Edition. For more information, see Configuring Destinations for SOAP.

For MDI SOAP Outbound (MDI to SAP S/4HANA Cloud Public Edition):

1. Create an MDI service instance for SAP S/4HANA Cloud Public Edition application.
2. Set up an MDI business system (one-time activity per tenant).
3. Set up outbound destinations on SAP BTP.
4. Set up communication arrangement SAP_COM_0008.
5. Subscribe to the Business Data Orchestration (BDO), and define a custom role collection. For more information, see Subscription Process for Business Data Orchestration.

6. Create a SOAP PUSH distribution model. For more information, see Configuring Distribution Models.

3.2.2 Setup Requirements for Other Business Objects Including Product

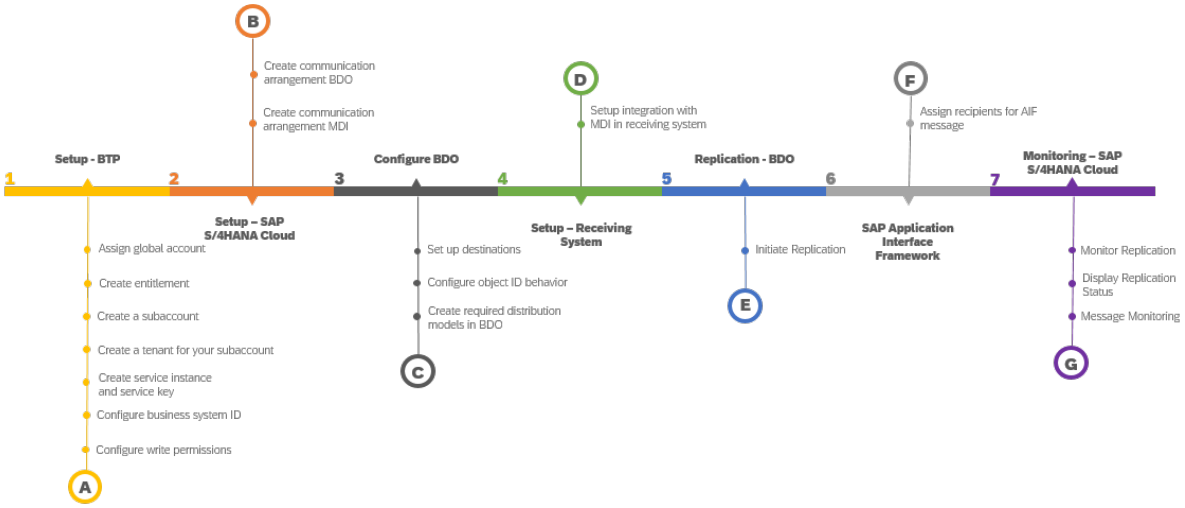
The setup for the other supported business objects, including Product, is a two-step process:

1. Set up MDI and S/4HANA Cloud Public Edition, and see the list of business objects that are supported.
2. Perform the setup between MDI and S/4HANA Cloud Private Edition, and see the list of business objects that are supported.

3.2.2.1 Data Replication Between MDI and SAP S/4HANA Cloud Public Edition

The following tasks let you replicate the object Product, as well as other business objects between MDI and SAP S/AHANA Cloud Public Edition.

The following diagram shows an overview of the data replication tasks:



To perform the required setup for the object Product, as well as other business objects, follow these steps:

- Follow the steps for the setup in SAP BTP, such as assigning a global account, and creating entitlement, among other steps.
- Follow the steps for the setup in SAP S/AHANA Cloud, such as creating communication arrangements.
- Configure Business Data Orchestration (BDO).
- Set up the integration with SAP MDI in the receiving system.
- Assign users in SAP application interface framework.
- Monitor messages and replication status in SAP S/AHANA Cloud.

For more information, see Master Data Replication Using SAP Master Data Integration.

Supported Master Data Objects for Replicating Data Between MDI and SAP S/4HANA Cloud Public Edition

The following business objects are supported in replicating data between MDI and SAP S/4HANA Cloud Public Edition:

SAP S/4HANA Cloud Public Edition as Provider and Consumer:

- Bank
- Business Partner
- Product
- Real Estate Rentable Object
- Technical Object
- Workforce Person

SAP S/4HANA Cloud Public Edition as Provider only:

- Company Code
- Cost Center
- Exchange Rate
- Plant
- Product Group
- Purchasing Organization
- Purchasing Organization Plant Assignment
- Real Estate Contract
- Real Estate Usable Object
- Symbolic General Ledger Account

SAP S/4HANA Cloud Public Edition as Consumer only:

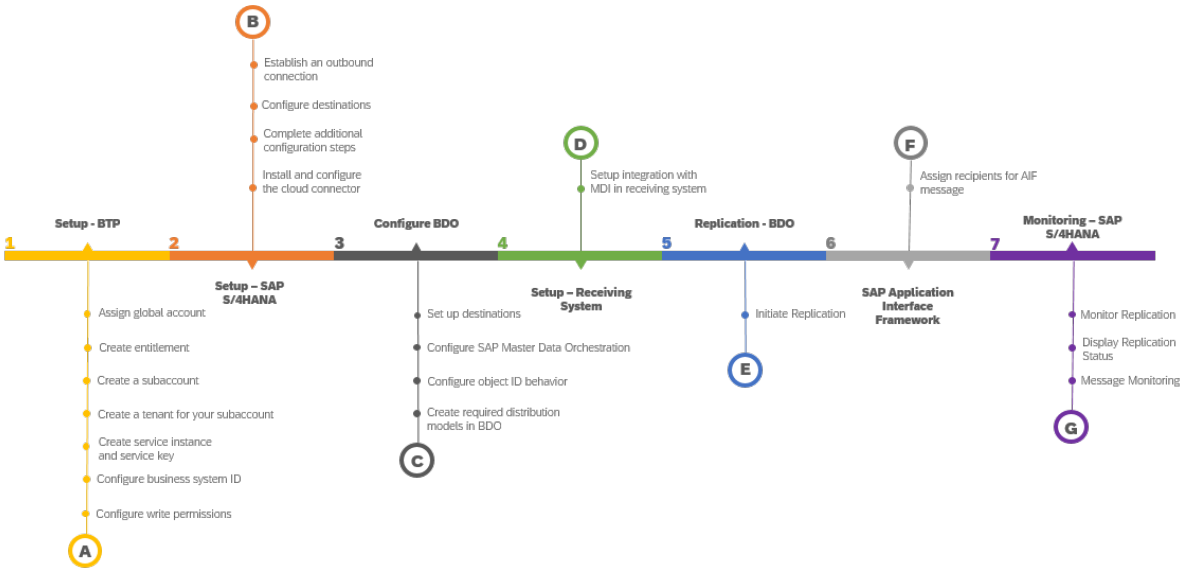
- Real Estate Architecture Object

For information about customizing specific master data, see Master Data-Specific Information.

3.2.2.2 Data Replication Between MDI and SAP S/4HANA Cloud Private Edition

Learn the required setup for specific business objects when you are replicating master data between MDI and SAP S/4HANA Cloud Private Edition.

The following diagram shows an overview of the required setup tasks for data replication:



- Follow the steps for the setup in SAP BTP, such as assigning a global account, and creating entitlement, among other steps.
- Follow the steps for the setup in SAP S/4HANA, such as establishing an outbound connection, among other steps.
- Configure Business Data Orchestration (BDO).
- Set up the integration with SAP MDI in the receiving system.
- Initiate the replication in BDO.
- Assign recipients to users for monitoring the messages in the SAP Application Interface Framework.
- Monitor messages and replication status in SAP S/4HANA.

For more information, see Master Data Replication: Setup Instructions for SAP S/4HANA.

Supported Master Data Objects for Replicating Data Between MDI and SAP S/4HANA Cloud Private Edition

The following business objects are supported in replicating data between MDI and SAP S/4HANA Cloud Private Edition:

SAP S/4HANA as Provider and Consumer:

- Bank
- Business Partner

- Product
- Real Estate Usable Object
- Real Estate Contract

SAP S/AHANA as Provider only:

- Cost Center
- Exchange Rate
- Plant
- Product Group
- Purchasing Organization
- Purchasing Group
- Public Sector Management Data
- Workforce
- Sales Controlling Object

SAP S/AHANA as Consumer only:

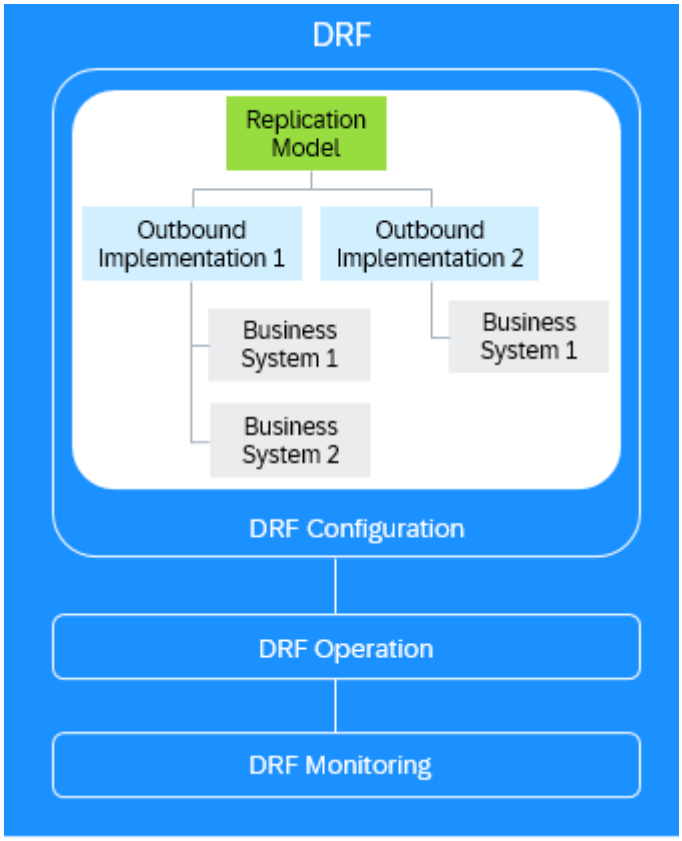
- Real Estate Architecture Object
- Real Estate Rentable Object

For information about customizing specific master data, see Master Data-Specific Information.

3.3 Data Replication Framework

Data Replication Framework (DRF) is one of two solutions for replicating your master data for implementing Two-Tier ERP.

DRF can be used to replicate master data such as business partners, products, and locations between the source and target systems. It is implemented in ABAP and is part of the SAP S/4HANA software stack. DRF always sends complete business object instances. The DRF is complemented by the key mapping framework and the value mapping framework to support non-harmonized identifiers and code lists.



3.3.1 Data Replication Framework Setup Requirements

The requirements for setting up master data replication using DRF depend on the system you are using: SAP S/4HANA Cloud Public Edition, or SAP S/4HANA.

DRF Setup from S/4HANA Cloud Public Edition

For setting up DRF from S/4HANA Cloud Public Edition, use the following communication scenarios:

- iSAP_COM_0008 – For Business Partner master data
- SAP_COM_0009 – For Product master data

DRF Setup from S/4HANA

In the transaction SOAMANAGER , perform the following tasks:

- Create domain systems.

- Create or assign a domain.
- Create Logon data.
- Create integration scenario configuration.

In the transaction DRFIMG, perform the following tasks:

- Define the technical settings for the business system.
- Define the replication model.

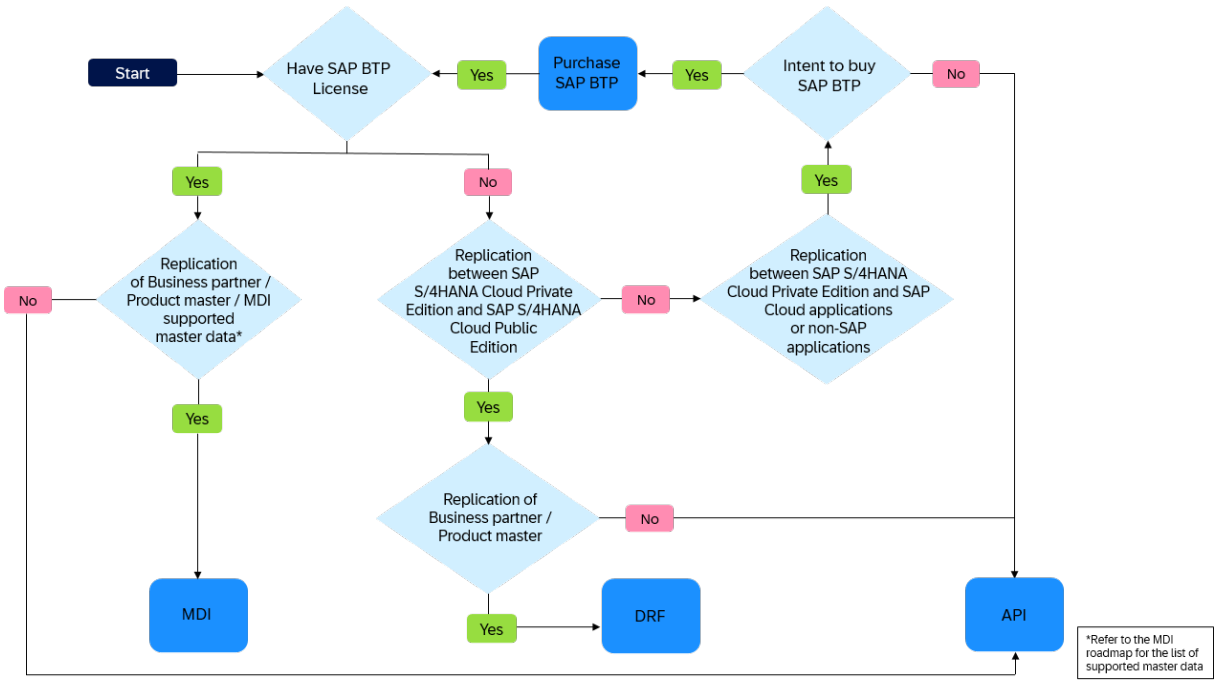
3.3.2 Supported Master Data Objects in Data Replication Framework

The following master data objects are supported in DRF:

- Customer
- Supplier
- Material Master

3.4 Master Data Replication Solutions

Use the following decision tree to help you arrive at the most optimal solution for you for replicating master data:



4 Transactional Data Replication

There are two options for replicating your transactional data:

- Cloud integration. For more information, see Cloud Integration for Replicating Transactional Data [page 16].
- Using API. For more information, see API for Replicating Transactional Data [page 18].

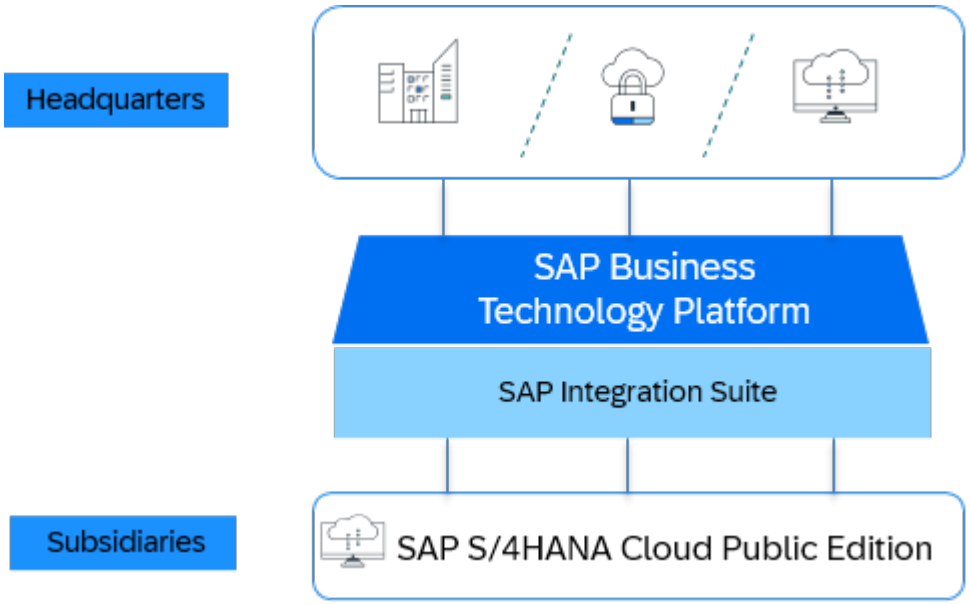
Choose the option that best suits your system landscape, and business requirements, among other factors.

4.1 Cloud Integration for Replicating Transactional Data

Cloud Integration (CI) is one of two solutions for replicating your transactional data when implementing Two-Tier ERP.

Consider opting for CI in the following cases:

- You have headquarters in private cloud and multiple subsidiaries in public cloud.
- You are using other SAP cloud solutions or non-SAP solutions that needed integration.
- Harmonizing master data across your system landscape is a requirement.
- You have a SAP Business Technology Platform account.

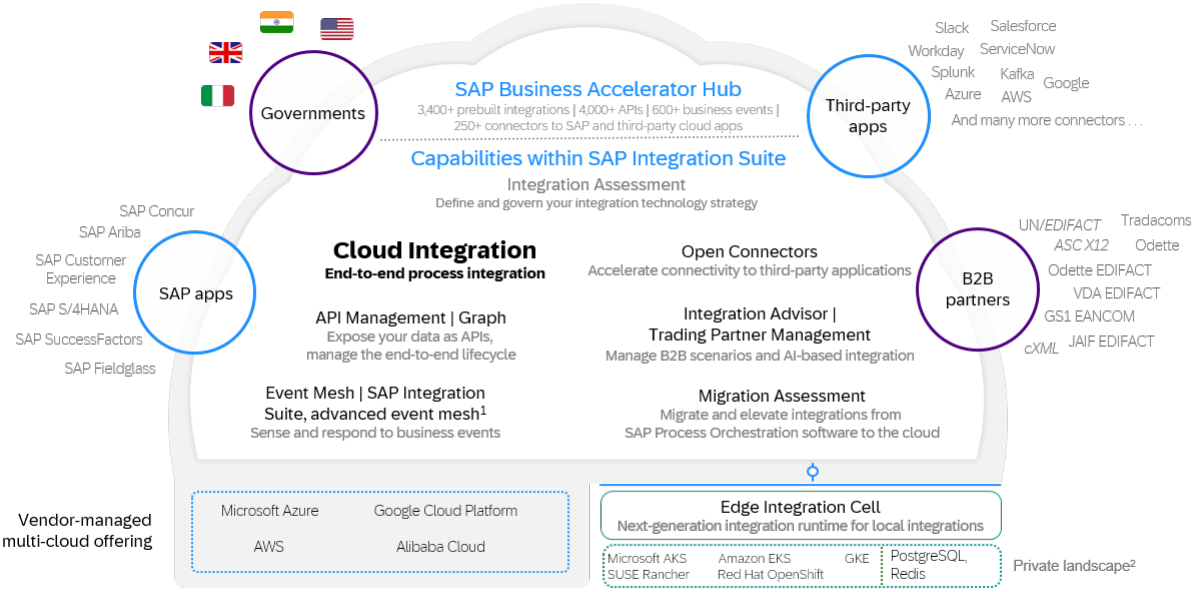


Cloud Integration

The SAP Cloud Integration service connects SAP S/4HANA private cloud edition and SAP S/4HANA public cloud edition with other SAP, non-SAP cloud, and on-premise applications.

This service is one of the capabilities of the SAP Integration Suite along with API Management, Integration Advisor, Open Connectors, and other integration capabilities for a cohesive and simplified toolkit for enterprise integration. To provide a comprehensive integration experience, these services are not available separately, but only as part of the Integration Suite service plan.

The following image shows the capabilities of the SAP Integration Suite, one of which is cloud integration:



[1] Optional separate license | [2] Private landscape corresponds to on premise or private cloud

For information about the SAP Integration Suite setup, see Initial Setup of SAP Integration Suite.

Here are the key aspects of Cloud Integration:

- It facilitates connections to both cloud and on-premise systems, whether they are SAP or non-SAP.
- It can handle real-time scenarios and offers a broad array of prebuilt integration content (Integration Packages) along with diverse capabilities and connectivity options.
- It provides you with iFlows that enable you to design message processing logic and configure sender/receiver adapters within SAP Cloud Integration. Prepackaged Two-Tier ERP iFlows include:
  - SAP S/4HANA Cloud for Sales Integration with SAP S/4HANA Cloud for Procurement or SAP ERP
  - SAP S/4HANA Cloud for Procurement Integration with SAP S/4HANA Cloud for Sales or SAP ERP

4.1.1 Cloud Integration Setup Requirements for Two-Tier ERP

Using Cloud Integration requires a series of tasks from SAP S/4HANA Cloud Public Edition and SAP S/4HANA Cloud Private Edition, and from the Cloud Integration service.

You must be subscribed to the SAP Integration Suite service plan to use the Cloud Integration service.

To start using Cloud Integration, do the following tasks from SAP S/4HANA Cloud Public Edition:

- Create communication user.
- Create communication system.
- Create communication arrangements.

From the Cloud Integration service, do the following tasks:

- Subscribe to and configure Cloud Integration instance for transactional data replication.
- Subscribe to and configure Process Runtime Integration.
- Generate Service Key and configure predefined iFlow based on scenarios.

From SAP S/4HANA Cloud Private Edition:

- Configure integration package.
- Configure the artifacts of the integration flow based on scenarios.

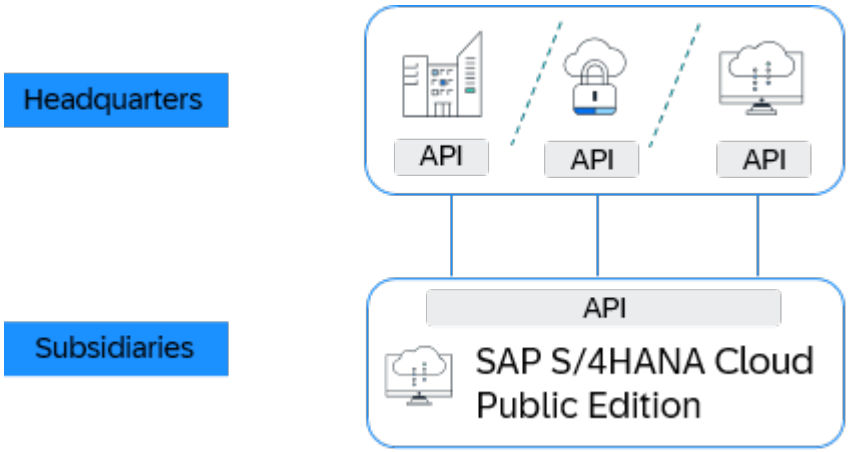
Related Information:

- SAP Cloud Integration.
- Cloud Integration - How to Setup Secure HTTP Inbound Connection with Client Certificates.
- Cloud Integration - How to Setup Secure Outbound HTTP Connection using Keystore Monitor.

4.2 API for Replicating Transactional Data

If you choose not to purchase a license for SAP Business Technology Platform but still need to fulfill the integration requirements, API Hub is the only viable option.

Here is an overview of a data integration solution using API:



The API Hub lets you:

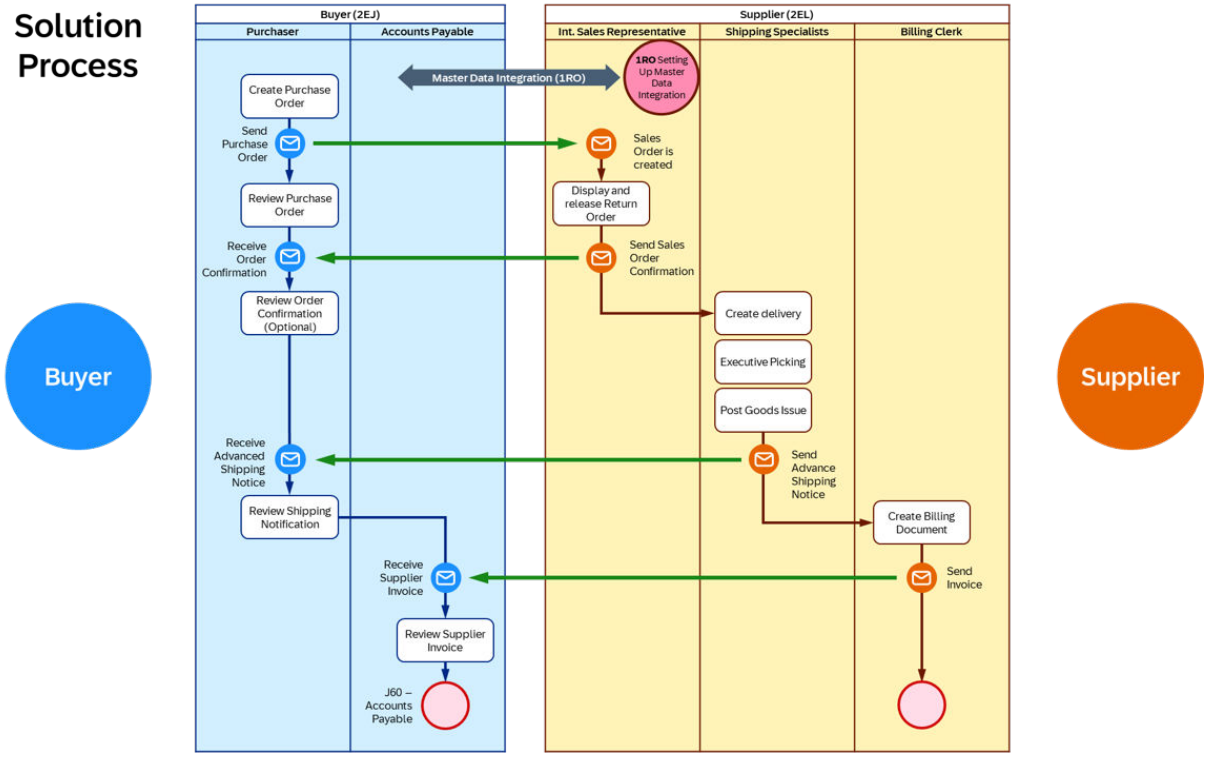
- Find content seamlessly to fit all your integration and extension needs.
- Explore different products, categories, business processes, partners, featured APIs, integrations, and more.
- Discover integrations between SAP products as well as between SAP products and third-party products.

- Use workspaces to start working on your content right away. Test APIs in both sandbox and configured systems.

5 Sample Integration Solution in a Drop Shipment Scenario

Find out the most suitable solution for replicating master data in a drop shipment scenario.

This graphic describes a typical drop shipment scenario:

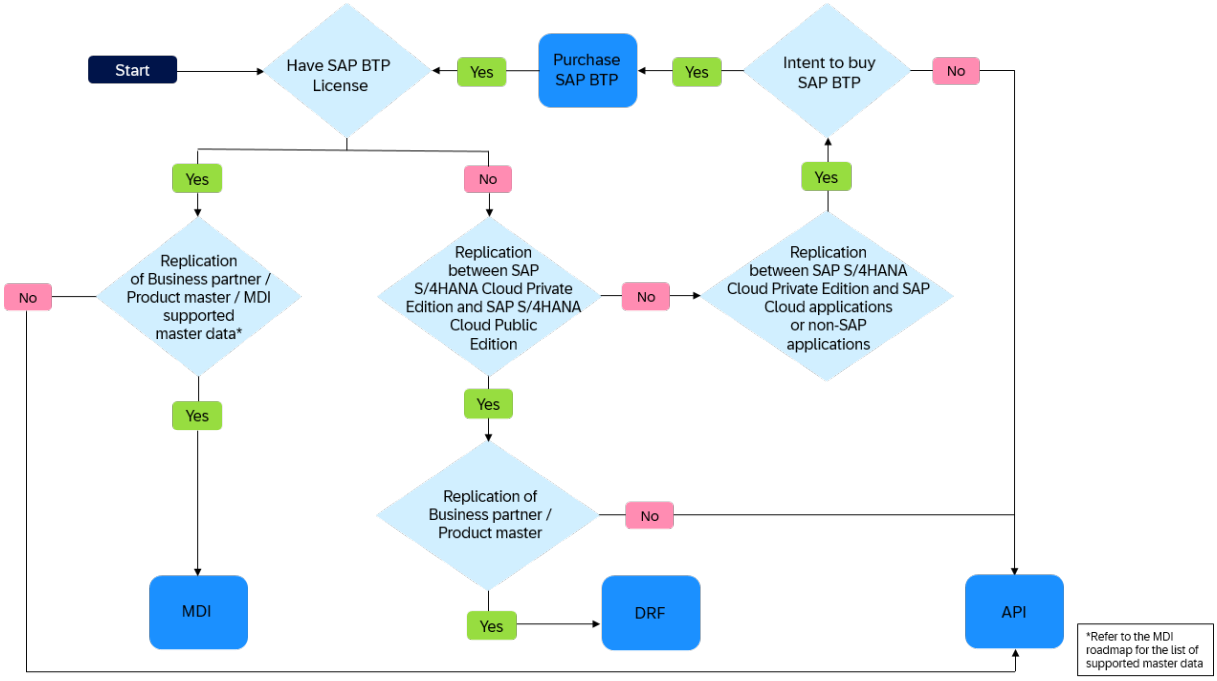


For example, the supplier, based in the headquarters, is on SAP S/4HANA Cloud Private Edition, and the buyer, a subsidiary, is on SAP S/4HANA Cloud Public Edition.

In the case of drop shipment, the following primary master data need to be replicated:

- Business Partner - Customer
- Business Partner - Supplier
- Product Master

Use this decision tree to determine which solution for master data replication fits the best:



Sample case

Have SAP BTP license?	Yes
Primary master data to replicate are Business Partner and Product Master	Yes

Based on the system landscape used and the master data that need to be replicated, the MDI is the best suited option.

6 Conclusion

The suggested ways of replicating the data are not exhaustive but represent the most relevant options, carefully selected based on their capabilities and alignment with customer needs. This list serves as a practical guide, offering the most suitable solutions tailored to address the specific requirements and objectives of different users. While there may be additional replication methods available, these recommendations prioritize effectiveness, scalability, and value in meeting the demands of real-world applications.

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